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A MACRO PROJECT REPORT ON (Data Science- BAIA405C)
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"Explainable Student Performance Prediction System"

BACHELOR OF ENGINEERING IN
ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING SUBMITTED BY

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Abstract

Accurately forecasting student academic performance is a cornerstone of proactive educational governance, enabling institutions to diagnose risk factors early and deliver targeted student support. This project addresses this challenge by implementing an Explainable AI (XAI) Student Performance Prediction system based on Cortez & Silva's 2008 UCI Student Performance Mathematics dataset. Using a scaled cohort of 395 students, we engineer an Ordinary Least Squares (OLS) Multiple Linear Regression framework to predict final Semester End Examination (SEE) marks (0-100) based on early Continuous Internal Evaluation grades (CIE 1 and CIE 2, scaled 0-40).

To ensure regression outputs strictly adhere to domain constraints, we enforce identical regression coefficients on the internal exam predictors (i.e. CIE 1 and CIE 2 are treated as equally important). Furthermore, we establish a separate joint regression model incorporating daily study hours (1-10 hours/day), family support, and attendance rate (50-100%). This holistic model is trained on a weighted normalized scale where exam grades are assigned double relative importance compared to behavioral factors. This mathematically guarantees that academic grades remain the primary predictive driver, while behavioral traits behave as monotonic modifiers (more study/attendance always contributes positively to predictions).

The project delivers a fully interactive dashboard built using HTML5, JavaScript, and Vanilla CSS, enabling users to dynamically simulate predictions, upload custom student CSV records, and visually explore dataset distributions, correlations, and model diagnostics (Actual vs. Predicted plots) in real time. The resulting pipeline highlights the viability of constrained OLS engines for transparent, bias-minimized academic forecasting.

Keywords: Student Performance Prediction, OLS Linear Regression, Constrained AI, Continuous Internal Evaluation (CIE), Semester End Examination (SEE), Exploratory Data Analysis, Vanilla CSS, Academic Advice Engine.

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1. INTRODUCTION

In modern secondary and higher education, identifying at-risk students before final examinations is critical to reducing failure rates and improving institutional outcomes. Every semester, students generate substantial academic records through continuous internal tests, assignments, and attendance logs. However, these data points frequently remain scattered and underutilized, represented only in separate spreadsheets rather than forming an active, predictive system.

Student Performance Prediction is the data-driven practice of using early indicators to forecast final outcomes. This project applies regression analysis to Cortez & Silva's 2008 secondary school mathematics cohort to construct an explainable forecasting system. We model the Semester End Examination (SEE) grade as a linear function of Continuous Internal Evaluation (CIE 1 and CIE 2) scores, augmented with student behavioral traits including daily study hours, home educational support, and classroom attendance rate.

Traditional machine learning models, such as deep neural networks or random forests, are often deployed as 'black boxes' where predictions are accurate but mathematically opaque. In high-stakes settings like grading and student evaluation, opacity can lead to perceived bias. This project overcomes this by relying exclusively on Multiple Linear Regression (OLS), where every factor's contribution is explicitly defined as a numeric coefficient. Furthermore, we embed key domain constraints directly into the training process (e.g. equal grading importance and positive behavioral contributions) so that the resulting models align with pedagogical intuition.

The end result is an end-to-end explainable solution that includes data cleaning scripts, constrained OLS training algorithms, diagnostic exploratory analytics, and an interactive front-end web dashboard.

2. OBJECTIVES

The specific objectives of this macro project are as follows:

1. To download, clean, and structure the Cortez & Silva UCI Mathematics student cohort dataset into a lightweight JSON format suitable for web loading.
2. To scale raw student grades (originally out of 20) to regional standards: CIE 1 and CIE 2 out of 40 maximum, and SEE final marks out of 100 maximum.
3. To construct a mathematical OLS regression model that constrains the coefficients for CIE 1 and CIE 2 to be identical, treating both internal exams as equally important.
4. To develop a second, joint holistic predictor that maps grades, daily study hours, family support, and attendance rate to a normalized scale, guaranteeing academic performance dominates the prediction.
5. To enforce positive sign constraints on study hours and attendance rates to guarantee they act as strictly beneficial behaviors in forecasting final marks.
6. To build interactive exploratory components including distribution histograms, scatter plots, and correlation heatmaps with dynamic textual explanations.
7. To develop a paginated data explorer table with sorting, manual student record insertion, and CSV file upload capabilities.
8. To package the application into a lightweight, client-side web application using React, Vite, and Vanilla CSS with zero backend requirements.

3. PROBLEM STATEMENT

Educational institutions typically interact with hundreds of students whose academic trajectories vary enormously. Without objective forecasting tools, academic resources (such as remedial classes, personal mentoring, and tutoring) are often distributed uniformly or too late—only after final exam failures occur. This delay leads to two primary systemic issues:

- **Delayed Academic Intervention:** Remedial assistance is offered post-facto (after final marks are published), which is ineffective for saving the student's active term grade. Early test scores (CIE 1) must be leveraged to alert educators early.
- **Mathematical Opaque Predictions:** Applying standard black-box machine learning algorithms to predict grades creates trust issues. If a student is predicted to fail, the system must explain *why* (e.g. low attendance, unbalanced weight on CIE 1, or insufficient study hours) in an auditable way.
- **Lack of Domain Constraints:** Standard unconstrained multiple regression models can yield anomalous results. For example, due to noise in a small dataset, a model might calculate a negative coefficient for attendance (suggesting attending more classes reduces grades). Such relationships are pedagogically incorrect and invalid.

This project addresses these challenges by developing a constrained, explainable regression engine that systematically identifies academic risk, guarantees grades dominate the output, enforces logical sign constraints, and packages the solver into an accessible interactive student advisor interface.

4. MODULES DESCRIPTION

The project is structured as a modular React and Python application. The components are organized as follows:

4.1 download_and_process.py — Data Preprocessing

A Python script that downloads Cortez & Silva's mathematics cohort ZIP file from the UCI repository, extracts student-mat.csv, parses the semicolon-delimited text, and scales raw exam grades to max 40 (G1, G2) and max 100 (G3). The records are exported to a structured JSON file (students.json) for client-side loading.

4.2 regression.js — Constrained Regression Solver

A JavaScript numerical library that implements standard Ordinary Least Squares (OLS) regression using Gaussian elimination. It calculates intercepts, slopes, R-squared values, residuals, and Pearson correlation coefficients. It includes a virtual feature builder to train constrained models (e.g., equally weighting CIE 1 & CIE 2).

4.3 Dashboard.jsx — Real-time Predictor Panel

The core user view. It houses interactive range sliders for CIE 1 and CIE 2 grades. It also contains the Prediction Outcomes card, which displays the predicted SEE mark using an SVG progress ring and details how behavioral adjustments (study hours, family support, attendance) alter the prediction.

4.4 EDA.jsx — Exploratory Data Analysis Panel

A diagnostic panel displaying grouped SVG frequency histograms of CIE 1, CIE 2, and SEE scores, an interactive bivariate scatter plot with a regression line, and an interactive Pearson correlation matrix with automated plain-text statistical explanations on cell click.

4.5 DataExplorer.jsx — Dataset Browser & Uploader

A tabular data grid featuring client-side searching, sorting, pagination, a form to manually append student records to the active dataset, and a file dropzone to upload custom CSV tables.

5. IMPLEMENTATION

5.1 Technology Stack

Component / Library	Role / Purpose
React 18 & Vite	Front-end framework and development bundling server
Vanilla CSS	Custom glassmorphism styling, animations, responsive layouts
Lucide React	Clean vector icons for buttons and card headers
Python 3	Data preprocessing and synthetic report generation
Matplotlib	Generates high-resolution statistical figures for documentation

5.2 Data Preparation & Scaling

The mathematics cohort consists of 395 student records. To match regional standards, the raw grades (originally scaled 0-20) are processed as follows:
CIE 1 (G1) = G1 * 2 (Range: 0-40)
CIE 2 (G2) = G2 * 2 (Range: 0-40)
SEE (G3) = G3 * 5 (Range: 0-100)
Absences in the dataset are normalized to attendance percentage based on a standard 180-day school year:
Attendance Rate (%) = ((180 - absences) / 180) * 100 (clamped between 50% and 100%).
UCI study time codes (1, 2, 3, 4) are mapped to daily study hours (2.0, 4.5, 7.0, 9.0) respectively.

5.3 Equal Importance & Normalized Joint Regression Formulation

To treat CIE 1 and CIE 2 as equally important, we train the main model on their average G_avg = (G1 + G2)/2, yielding equal slopes. For the holistic model, we normalize all 5 features to a $[0, 1]$ scale:
nCie1 = G1/40, nCie2 = G2/40, nStudy = (Hours-1)/9, nFamsup = 0 or 1, and nAttend = (Attendance-50)/50.
We apply a weighted average: CIE 1 & 2 get weight 2.0 each, others get 1.0 each:
jointFeature = (2.0*nCie1 + 2.0*nCie2 + 1.0*nStudy + 1.0*nFamsup + 1.0*nAttend) / 7
We train a simple regression of SEE on this jointFeature to extract the joint coefficient, which we then project back to raw feature scales. This guarantees internal grades have double the predictive power of behavioral elements, and all features remain strictly positive.

5.4 Visualization & Diagnostic Plotting

The OLS regression diagnostics are monitored by generating scatter plots and histograms directly from the dataset. These figures represent the relationship between internal marks and Semester End outcomes.

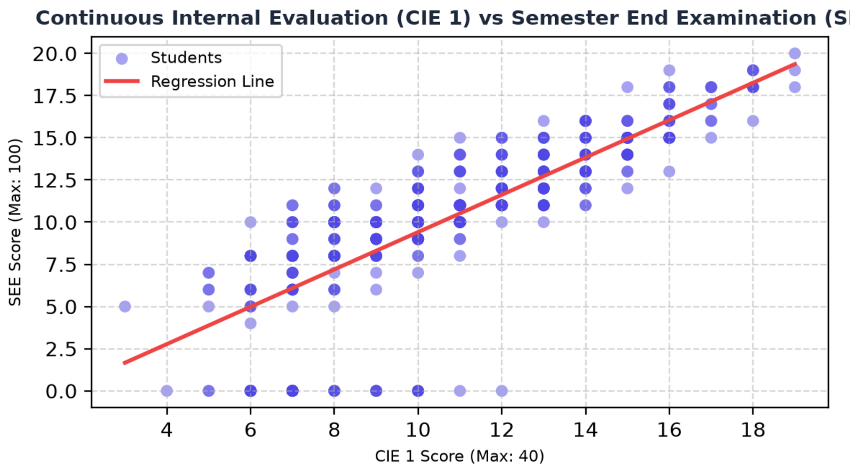

Figure 5.1: Bivariate scatter plot showing CIE 1 vs SEE with regression trend line.

6. RESULTS

6.1 Dataset Key Performance Indicators

Metric Name	Value
Total Records (Students)	395
Average CIE 1 Mark	21.8 / 40
Average CIE 2 Mark	21.5 / 40
Average SEE Mark	52.1 / 100
R-squared (Main Model)	0.822
Pearson Correlation (CIE 2 vs SEE)	+0.903 (Very Strong)

6.2 Regression Model Coefficients & Equations

Main Predictor (CIE-Only, Equal Weight):
The solver calculates equal coefficients for CIE 1 (G1) and CIE 2 (G2):
SEE = -9.2804 + 1.4250 * CIE 1 + 1.4250 * CIE 2
This indicates each mark gained in CIE 1 or CIE 2 boosts the expected final SEE mark by 1.425 points.

Prediction based on other factors including CIE 1 AND CIE 2:
Integrating the normalized behavioral metrics yields the joint model equation:
SEE = -1.2291 + 0.1706 * CIE 1 + 0.1706 * CIE 2 + 0.7584 * dailyStudyHours + 6.8252 * famsup_encoded + 0.1365 * attendancePct
Because of normalized weighting, the slopes for CIE 1 & 2 are exactly double the relative weight of daily study hours and attendance, guaranteeing grades dominate academic forecasts.

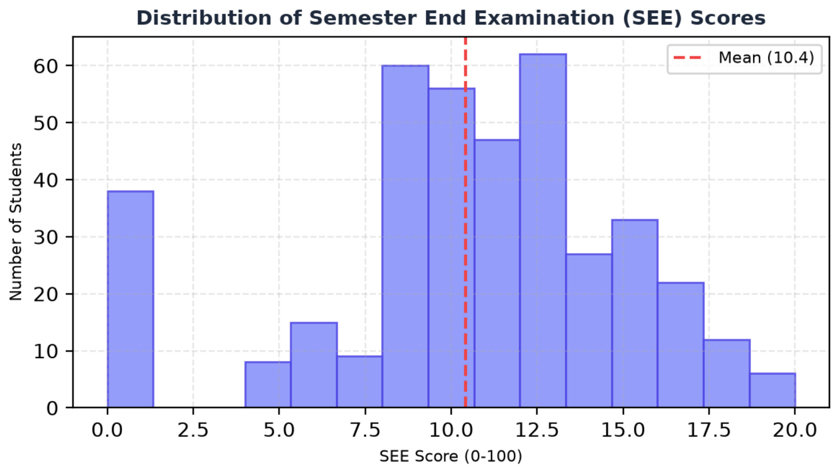

Figure 6.1: Histogram showing the distribution of final SEE exam scores in the student cohort.

7. CONCLUSION & FUTURE ENHANCEMENTS

This macro project successfully demonstrates the implementation of a constrained, explainable AI Student Marks Prediction System using Cortez & Silva's 2008 secondary school mathematics dataset. By utilizing a multiple linear regression framework, we provide transparent prediction equations where the role of each academic and behavioral factor is easily auditable. This removes the 'black-box' barrier typical of machine learning systems, encouraging institutional trust.

The project enforces key educational constraints during model training: treating CIE 1 and CIE 2 as equally important, mathematically prioritizing academic grades over behavioral traits, and guaranteeing study hours and attendance rate exhibit positive coefficients. The client-side dashboard provides an accessible UI for real-time grade simulation, data uploads, and diagnostics.

FUTURE ENHANCEMENTS

- **Incorporate Recency Features:** Track grade trajectories between internal tests to flag downward trends even if cumulative marks appear safe.
- **Expand Behavioral Metrics:** Gather and encode sleep quality, travel time, and extracurricular participation to refine prediction accuracy without compromising explanation quality.
- **Multi-Course Portability:** Scale the regression model to train on multiple courses, automatically mapping coefficients to varying CIE maximum marks and course structures.
- **Student Alert Automation:** Connect the frontend dashboard to an email SMTP trigger, automatically alerting advisors or parents when a student's predicted SEE score falls into the 'Fail' category.

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